

SIMATIC

ET 200SP HA Digital Input Module DI 32x24VDC HA (6DL1131-6BL00-0PH1)

Equipment Manual

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Legal information

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This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

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Preface

Validity of the documentation

This manual describes the following I/O module:

System family	I/O module	Article No.
SIMATIC; Distributed I/O System; ET 200SP HA	DI 32x24VDC HA	6DL1131-6BL00-0PH1

It supplements the system manual for the ET 200SP HA distributed I/O system.

Functions that generally relate to the system are described in this manual.

The information in this device manual and the system/function manuals enables you to put the ET 200SP HA into operation.

Configuration of the ET 200SP HA with PCS 7 (STEP 7)

You can configure the following components of the ET 200SP HA when the following updates are installed:

Component of the ET 200SP HA	Required updates
DI 32x24VDC HA	HSP0275

Appendices

The appendices provide information that is relevant for using the ET 200SP HA outside the PCS 7 environment.

Conventions

Please also observe notes marked as follows:

Note

A note contains important information on the product described in the documentation, on the handling of the product or on the section of the documentation to which particular attention should be paid.

Product overview

3.1 Properties of the I/O module DI 32x24VDC HA

View of the module

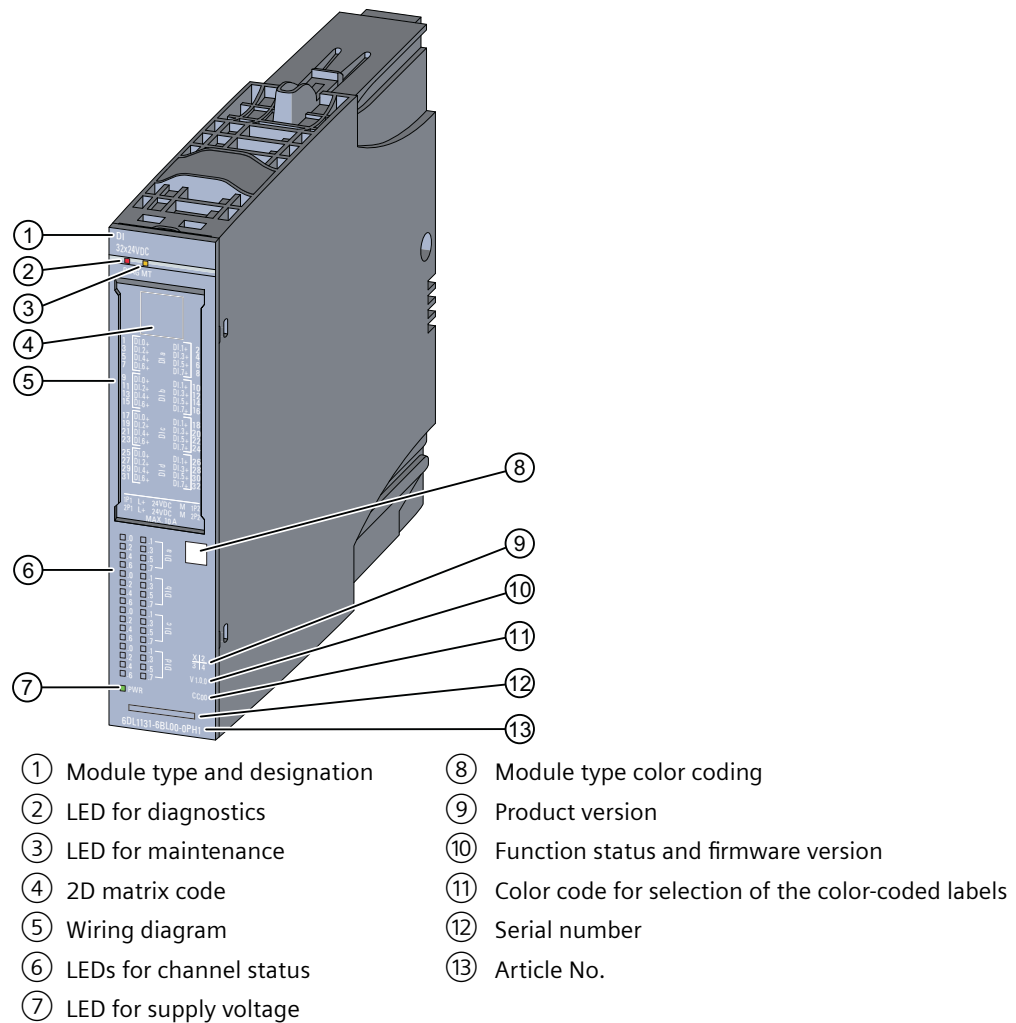


Figure 3-1 View of the DI 32x24VDC HA

Properties

The I/O module has the following technical properties:

- 32 digital inputs with the following properties that can be individually configured for each channel
 - Diagnostics (wire break)
 - Hardware interrupts for positive and negative edges
- Sink input (PNP, active high)
- Module-specific configurable diagnostics for missing supply voltage L+
- Suitable for connecting switches and 2-wire sensors according to IEC 61131-2, types 1 and 3

The I/O module supports the following functions:

- Firmware update
- I&M identification data
- Parameter reassignment in RUN
- Value status QI

You can configure the I/O module with HW Config and integrate it into your system.

Accessories

The following accessories must be ordered separately:

- Labeling strips
- Color-coded labels
- Reference identification labels
- Shield connector

Wiring

4.1 Terminal blocks for the I/O module

You can operate the I/O module with the following terminal blocks:

- 6DL1193-6TP00-0DH1, light
- 6DL1193-6TP00-0BH1, dark
- 6DL1193-6TP00-0DM1, light, for redundant configuration
- 6DL1193-6TP00-0BM1, dark, for redundant configuration
- 6DL1193-6TP00-0DN0, light, terminal block with potential distributor on the right side for connecting actuators to ground
- 6DL1193-6TP00-0BN0, dark, terminal block with potential distributor on the right side for connecting actuators to ground
- 6DL1193-6TP00-0DP0, light, terminal block with potential distributor on the right side for encoder supply
- 6DL1193-6TP00-0BP0, dark, terminal block with potential distributor on the right side for encoder supply

The terminal blocks are not included in the scope of delivery of the I/O module and must be ordered separately.

Note

You can find additional information on the configuration in the system manual.

General pin assignment

Terminal	Assignment	Terminal	Assignment	Explanations
1	DI a /DI.0+	2	DI a /DI.1+	DI a /DI.n+: Input signal, channel n (0...7)
3	DI a /DI.2+	4	DI a /DI.3+	DI b /DI.n+: Input signal, channel n+8 (8...15)
5	DI a /DI.4+	6	DI a /DI.5+	DI c /DI.n+: Input signal, channel n+16 (16...23)
7	DI a /DI.6+	8	DI a /DI.7+	DI d /DI.n+: Input signal, channel n+24 (24...31)
9	DI b /DI.0+	10	DI b /DI.1+	1P1: Supply voltage L+ of the voltage bus 1P
11	DI b /DI.2+	12	DI b /DI.3+	2P1: Supply voltage L+ of the voltage bus 2P
13	DI b /DI.4+	14	DI b /DI.5+	1P2: Ground reference of the voltage bus 1P
15	DI b /DI.6+	16	DI b /DI.7+	2P2: Ground reference of the voltage bus 2P
17	DI c /DI.0+	18	DI c /DI.1+	
19	DI c /DI.2+	20	DI c /DI.3+	1 $\overline{\text{DI.0+}}$ 2
21	DI c /DI.4+	22	DI c /DI.5+	3 $\overline{\text{DI.2+}}$ 4
23	DI c /DI.6+	24	DI c /DI.7+	5 $\overline{\text{DI.4+}}$ 6
25	DI d /DI.0+	26	DI d /DI.1+	7 $\overline{\text{DI.6+}}$ 8
27	DI d /DI.2+	28	DI d /DI.3+	9 $\overline{\text{DI.0+}}$ 10
29	DI d /DI.4+	30	DI d /DI.5+	11 $\overline{\text{DI.2+}}$ 12
31	DI d /DI.6+	32	DI d /DI.7+	13 $\overline{\text{DI.4+}}$ 14
1P1	L+	1P2	M	15 $\overline{\text{DI.6+}}$ 16
2P1	L+	2P2	M	17 $\overline{\text{DI.0+}}$ 18
				19 $\overline{\text{DI.2+}}$ 20
				21 $\overline{\text{DI.4+}}$ 22
				23 $\overline{\text{DI.6+}}$ 24
				25 $\overline{\text{DI.0+}}$ 26
				27 $\overline{\text{DI.2+}}$ 28
				29 $\overline{\text{DI.4+}}$ 30
				31 $\overline{\text{DI.6+}}$ 32
				1P1 L+ 24VDC M 1P2
				2P1 L+ 24VDC M 2P2
				MAX. 10 A
				Pin assignment on the front of the module

Connection scheme	Note
DI.n+ ← DC 24 V ←	For example, when using an external power supply with ground connection

4.3 Pin assignment of the I/O module DI 32x24VDC HA (2-wire connector)

General pin assignment with potential distributor

- The pin assignment on the left side of the terminal blocks with potential distributor corresponds to the assignment in the section "Pin assignment of the I/O module DI 32x24VDC HA (1-wire connector) (Page 12)".
- The potential distributor with the following terminals is located on the right side of the terminal block:
 - 32 L+ terminals for power supply of sensors.
 - Four M terminals

Note

- The L+ terminals on the potential distributor are intended exclusively for the sensors connected to the I/O module using a 2-wire connection system.
- You may not use these terminals as voltage source for other devices in the system.

Recommendation: Use terminal blocks with potential distributor

For the connection of sensors with two wires, the following terminal block should be used:

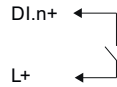
Article No.	Function/parameter
6DL1193-6TP00-ODP0	Allow new potential group (light terminal block)
6DL1193-6TP00-OBP0	Use potential group of the left module (dark terminal block)

Table 4-2 Benefits of potential distributors compared to using external power supply units

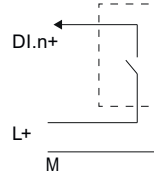
Advantage with terminal block with potential distributor	Disadvantage without potential distributor
The L+ terminals on the potential distributor are connected individually using automatic circuit breakers to a maximum of 20 mA. In the event of a short-circuit or overload, "only" the affected channel fails. A wire break is signaled if the parameters "Diagnostics, Wire break" and "Check for wire break" are enabled.	No separate protection Short circuit or overload affects the entire system.

Connection of sensors with potential distributor or external power supply

The following figures show the connection of sensors with potential distributor or external power supply.



Sensor connection DI - 2-wire



Sensor connection with external power supply

These connection variants are approved only with terminal blocks of the type "potential distributor".

Additional information

You can find additional information on the configuration in the system manual *SIMATIC; Distributed I/O System; ET 200SP HA*.

4.4 Schematic circuit diagram

The following figure shows the block diagram of the DI 32x24VDC HA.

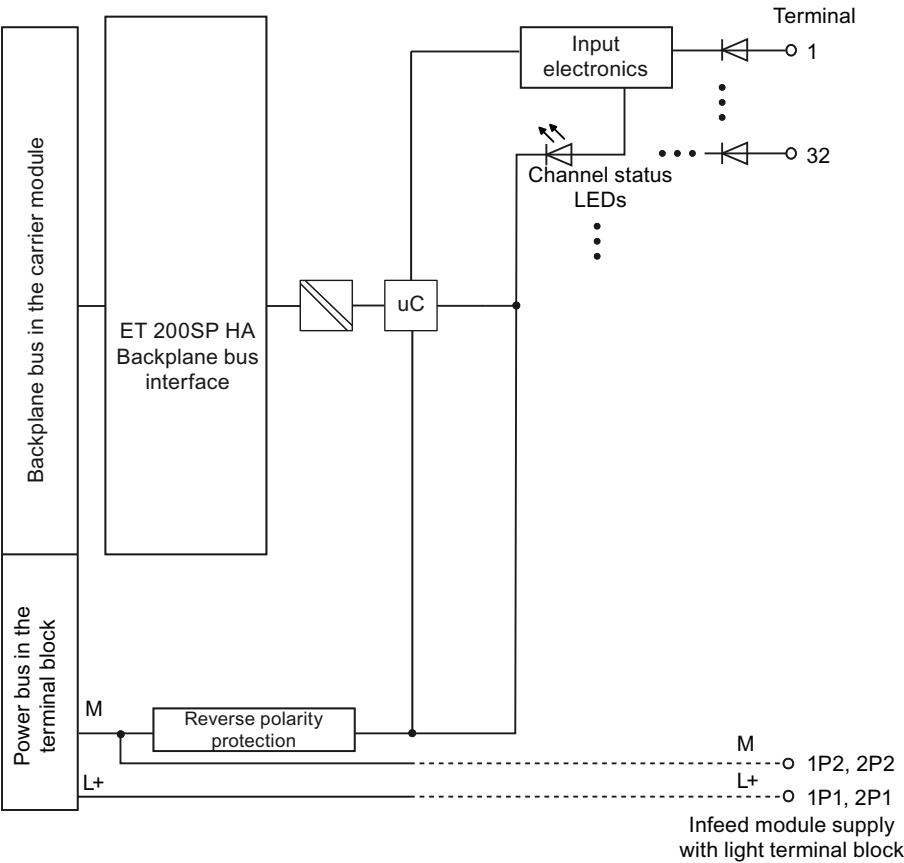


Figure 4-1 Block diagram DI 32x24VDC HA

Parameters

5.1 Parameters of the I/O module DI 32x24VDC HA

Configuration

You configure the I/O module with PCS 7 V9.0 or higher.

Parameters

You define the functioning of the I/O module via parameters.

See also

Parameter assignment and structure of the module/channel parameters (Page 32)

5.2 Module/channel parameters

Parameters of the DI 32x24V DC HA

You have the following setting options:

Table 5-1 Parameters and their defaults

Parameter	Value range	Default	Parameter re-assignment in RUN	Efficiency range
Diagnostics, Missing supply voltage L+	- Enabled - Disabled	Enabled	Yes	Module
Diagnostics, Wire break ¹	- Enabled - Disabled	Enabled	Yes	Channel
Check for wire break	- Enabled - Disabled	Enabled	Yes	Channel
Channel activated	- Enabled - Disabled	Enabled	Yes	Channel
Hardware interrupt, rising edge	- Enabled - Disabled	Disabled	Yes	Channel
Hardware interrupt, falling edge	- Enabled - Disabled	Disabled	Yes	Channel
Potential group	- Use potential group of the left module - Allow new potential group	Use potential group of the left module	No	Module
Double-wide terminal block ²	- Enabled - Disabled	Disabled	No	Configuration (HW Config)

¹ The wire break diagnostics can only be enabled when the "Check for wire break" parameter is enabled. If you use a simple encoder contact, you need to connect a parallel resistor to enable wire break diagnostics in the open state (sensor resistance for wire break diagnostics: 15 kΩ to 18 kΩ).

² I/O module on terminal block with potential distributor: Select the "Double-width terminal block" parameter in the hardware properties for the I/O module.

5.3 Explanation of the module/channel parameters

Diagnostics, Missing supply voltage L+

Enabling of the diagnostics for missing or insufficient supply voltage L+.

Diagnostics, Wire break

When wire-break check is activated, you can enable wire-break diagnostics via the Diagnostics, Wire break parameter for insufficient current flow on the input.

Check for wire break

Enable for wire break check. The module can only detect a wire break when the wire-break check is activated.

Channel activated

Specifies whether a channel is activated or deactivated.

Hardware interrupts positive edge

Specifies if a hardware interrupt should be generated for a rising edge.

Hardware interrupts negative edge

Specifies if a hardware interrupt should be generated for a falling edge.

Potential group

A potential group consists of a group of terminal blocks arranged next to one another, with the associated I/O modules within an ET 200SP HA station.

Starting a potential group

- The first (left-hand) terminal block which you install in the IO device (on the first carrier module to the right next to the interface module) is a light-colored terminal block.
- Connect the supply voltage to a light-colored terminal block.
- Each light-colored terminal block which you install in the IO device starts a new potential group.
- The power bus is interrupted to the left.

Extending the potential group

- Each dark-colored terminal block passes the supply voltage connected on the light-colored terminal block to the I/O modules and to the dark terminal block positioned to the right of it.
- The power bus is continuous.

5.3 Explanation of the module/channel parameters

End of a potential group

- A potential group ends with the dark terminal block, which follows a light terminal block or a server module in the station configuration.

Power supply of I/O modules of a potential group

- All I/O modules of a potential group have a shared power supply.

You can find additional information on the configuration of the potential group in the system manual *SIMATIC; Distributed I/O System; ET 200SP HA*.

Double-width terminal block (I/O module on terminal block with potential distributor)

Select the "Double-width terminal block" parameter in the hardware properties for the I/O module.

The parameter only has an effect on the configuration in HW Config.

Displays and interrupts

6.1 Status and error displays of the I/O module DI 32x24VDC HA

LED displays

The following figure shows the LED displays of the I/O module:

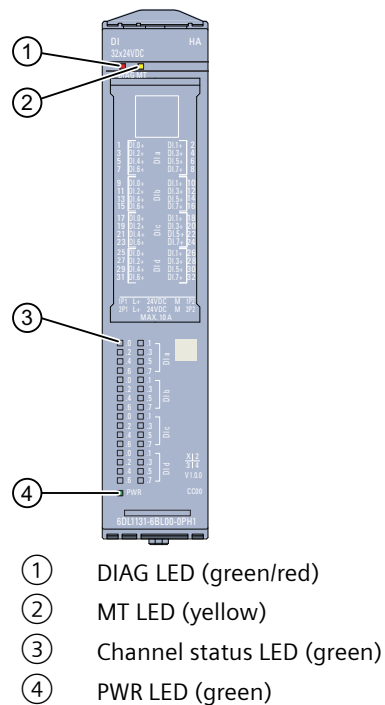


Figure 6-1 LED displays

Meaning of the LED displays

The tables below explain the meaning of the status and error displays. Remedial measures for diagnostics alarms can be found in section Diagnostics messages and maintenance events (Page 34).

DIAG LED

Table 6-1 Diagnostics display of the DIAG LED

DIAG LED	Meaning
 Off	The power supply of the ET 200SP HA is disrupted or switched off.
 Flashes	Module is not configured.
 On	Module is configured, a diagnostics message is not pending.
 Flashes	Module is configured, at least one diagnostic message is pending.

MT LED

Table 6-2 Status display of the LED MT

MT LED	Meaning
 Off	No maintenance required.
 On	Maintenance is required, i.e. at least one maintenance event has occurred.

Channel status LED

Regardless of the parameter assignment, the channel status LED status shows the signal level at the input terminal. It lights up starting at approximately 10 V of terminal voltage.

Table 6-3 Status display of the channel status LEDs

LED channel status	Meaning
 Off	Process signal = 0
 On	Process signal = 1

PWR LED

Table 6-4 Status display of the PWR LED

PWR LED	Meaning
 Off	Supply voltage L+ missing.
 On	Supply voltage L+ present.

6.2 Interrupts

The I/O module supports diagnostics interrupts, maintenance messages and hardware interrupts. You can find additional information on hardware interrupts in the appendix Hardware interrupts (Page 36)

Diagnostics interrupts

The I/O module generates a diagnostics interrupt at the following events:

- Supply voltage L+ missing
- Parameter assignment error
- Hardware interrupt lost
- Channel/component temporarily unavailable
- Wire break
- Retentive memory in carrier module defective
- Retentive memory in the terminal block defective
- Module is faulty

Maintenance messages

The I/O module generates a maintenance message for the following events:

- Retentive memory in carrier module defective
- Retentive memory in the terminal block defective

Technical specifications

Technical specifications of the DI 32x24VDC HA

Article number	6DL1131-6BL00-0PH1
General information	
Product type designation	DI 32x24VDC HA
Firmware version	V1.0
• FW update possible	Yes
Usable terminal block	TB type H1 and P0
Color code for module-specific color identification plate	CC00
Product function	
• I&M data	Yes; I&M0 to I&M3
Engineering with	
• PCS 7 configurable/integrated as of version	V9.0
Operating mode	
• DI	Yes
• Counter	No
• Oversampling	No
• MSI	No
Redundancy	
• Redundancy capability	No
Supply voltage	
Rated value (DC)	24 V
permissible range, lower limit (DC)	19.2 V
permissible range, upper limit (DC)	28.8 V
Reverse polarity protection	Yes
Input current	
Current consumption (rated value)	11.5 mA
Current consumption, max.	15 mA
Encoder supply	
Number of outputs	32; TB45-P32+32x24V+4M
Output voltage encoder supply, min.	19.2 V
Short-circuit protection	Yes; When using TB type P0
Address area	
Address space per module	
• Address space per module, max.	4 byte; + 4 bytes for QI information
Hardware configuration	
Automatic encoding	

Article number	6DL1131-6BL00-0PH1
Mechanical coding element	Yes
Digital inputs	
Number of digital inputs	32
Digital inputs, parameterizable	Yes
Source/sink input	Yes; P-reading
Input characteristic curve in accordance with IEC 61131, type 1	Yes
Input characteristic curve in accordance with IEC 61131, type 2	No
Input characteristic curve in accordance with IEC 61131, type 3	Yes
Pulse extension	No
Edge evaluation	Yes; rising edge, falling edge, edge change
Input voltage	
Type of input voltage	DC
Rated value (DC)	24 V
for signal "0"	-30 to +5V
for signal "1"	+11 to +30V
Input current	
for signal "1", typ.	2.5 mA
Input delay (for rated value of input voltage) for standard inputs	
parameterizable	No
at "0" to "1", min.	3.2 ms
at "0" to "1", max.	5.3 ms
at "1" to "0", min.	2.9 ms
at "1" to "0", max.	4.5 ms
Cable length	
shielded, max.	1 000 m
unshielded, max.	600 m
Encoder	
Connectable encoders	
2-wire sensor	Yes
permissible quiescent current (2-wire sensor), max.	1.5 mA
Interrupts/diagnostics/status information	
Diagnostics function	Yes
Alarms	
Diagnostic alarm	Yes; channel by channel
Hardware interrupt	Yes; channel by channel
Diagnostic messages	
Diagnostic information readable	Yes
Monitoring the supply voltage	Yes; Module-wise

Article number	6DL1131-6BL00-0PH1
– parameterizable • Wire-break • Short-circuit	Yes Yes; Channel-by-channel, optional protective circuit for preventing wire-break diagnostics in the case of simple encoder contacts: 15 kOhm to 18 kOhm No
Diagnostics indication LED	
• MAINT LED • Monitoring of the supply voltage (PWR-LED) • Channel status display • for channel diagnostics • for module diagnostics	Yes; yellow LED Yes; green PWR LED Yes; Green LED No Yes; green/red DIAG LED
Potential separation	
Potential separation channels	
• between the channels • between the channels and backplane bus • between the channels and the power supply of the electronics	No Yes No
Isolation	
Isolation tested with	1 500 V DC/1 min, type test
Ambient conditions	
Ambient temperature during operation	
• horizontal installation, min. • horizontal installation, max. • vertical installation, min. • vertical installation, max.	-40 °C 70 °C -40 °C 60 °C
Dimensions	
Width	22.5 mm
Height	115 mm
Depth	138 mm
Weights	
Weight, approx.	150 g

Cycle time

The cycle time describes the time slice in which signals from the inputs are acquired and processed. The cycle time of the I/O module is 250 µs.

Minimum pulse length

In order for a pulse or an edge transition to be reliably detected by the I/O module, the signal state must be active for at least the following duration:

Input delay + cycle time

Drivers, parameters, diagnostics messages and address space

A

A.1 Concept of the driver and diagnostics blocks in PCS 7

Use with PCS 7

The following information shows you the benefits of using PCS 7. PCS 7 automatically carries out many important parameterizations and interconnections of blocks.

If you want to use the ET 200SP HA outside of the PCS 7 environment, note the next, following section.

Tasks of the driver and diagnostics blocks (driver blocks)

In process control systems, certain demands are placed on the diagnostics/signal processing. This includes the monitoring of modules, IO devices and IO controllers for malfunctions and failures.

To enable this, blocks are available in the PCS 7 library that implement the interface to the hardware including test functions.

These blocks perform two basic tasks:

- They provide signals from the process to the AS for further processing.
- They monitor modules, IO devices and IO controllers for failure.

When the process signals are read in, these blocks access the process image input (or process image partition) (PII) and when the process signals are output, they access the process image output (or process image partition) (PIQ).

Concept

The concept of the driver and diagnostics blocks for PCS 7 can be characterized as follows:

- The separation between user data processing (CHANNEL blocks) and diagnostics data processing (MODULE blocks)
- The symbolic addressing of the I/O signals
- The automatic generation of the MODULE blocks by CFC

A.1 Concept of the driver and diagnostics blocks in PCS 7

This block concept supports all modules from the list of approved modules.

When new Siemens or non-Siemens module types are integrated, the meta-knowledge for the driver generator can be extended by additional XML files (object and action lists).

Note

Driver and diagnostics blocks

- The library with the driver blocks has to be installed using the Setup program on the PC. This is the only method of ensuring that the meta-knowledge required for the driver generator is available. You must not copy the library from another computer.
 - You can also use driver blocks from another library (for example, your own blocks from your own library). You can specify this additional library in the "Generate module drivers" dialog box. The driver generator then searches for the block to be imported in the library specified here. If the block is not found here, it is searched for in the library specified in the control file (XML file).
 - If the S7 program contains a signal-processing block but not from one of the PCS 7 libraries, you have to specify the version of the driver library from which the driver blocks are to be imported in the "Generate module drivers" dialog box.
-

Time-optimized processing

For time-optimized processing of the user program during runtime, PCS 7 automatically carries out the following programming tasks:

- Divide organization blocks for error handling (for example, OB 85, OB 86) into runtime groups
- Installing driver blocks into the corresponding runtime groups

If an error occurs, the SUBNET block, for example, activates the relevant runtime group, the RACK block or MODULE block contained in the runtime group detects the error, evaluates it and outputs a process control message to the OS.

The diagnostics information of the MODULE block (OMODE_xx output) is also transferred to the corresponding CHANNEL block (MODE input).

If necessary, this information can be displayed by means of a PCS 7 block that can be operated and monitored on the OS or by means of a user block in a process picture (color change of the measured value or flashing display, etc.).

A.2 Parameters, diagnostics messages and address space without PCS 7

A.2.1 Parameter assignment

Parameter assignment in the user program

You can reconfigure individual channels of the module in RUN without this having an effect on the other channels.

Changing parameters in RUN

The parameters are transferred to the module with the "WRREC" instruction via data record 128.

The parameters set in STEP 7 are not changed in the CPU, which means that the parameters set in STEP 7 will be valid again after a restart of the CPU.

If you reconfigure a module and diagnostics are pending prior to the reconfiguration, these diagnostics are not signaled as "outgoing".

Output parameter STATUS of the WRREC

If errors occur when transferring parameters with the "WRREC" instruction, the module continues operation with the previous parameter assignment. The STATUS output parameter contains the corresponding error code.

The STATUS output parameter is 4 bytes long and is configured as followed:

- Byte1: Function_Num, general error code
- Byte2: Error_Decode, location of the error code
- Byte3: Error_Code_1, error code
- Byte4: Error_Code_2, manufacturer-specific extension of the error code

Module-specific errors are displayed via Error_Decode = 0x80 and Error_Code_1 / Error_Code_2.

Error_Code_1	Error_Code_2	Cause	Remedy
0xB0	0x00	Number of the data record unknown	Enter valid number for data record.
0xB1	0x01	Length of the data record is incorrect	Enter valid value for data record length.
0xB2	0x00 to 0xFF	Module cannot be reached	• Check station - is the module plugged in correctly? • Check the parameters of the WRREC instruction.
0xE0	0x01	Wrong version in the header	Correct the version number of the parameter block.
0xE0	0x02	Header error (number or length of parameter block)	Correct the length and number of the parameter blocks.

Error_Code_1	Error_Code_2	Cause	Remedy
0xE1	0x01	Reserved bit set	Check and correct the parameters
0xE1	0x02	Invalid diagnostics enable bit set for operating mode	
0xE1	0x0E	Invalid redundancy parameter assignment	

A.2.2 Parameter assignment and structure of the module/channel parameters

Structure of data record 128

Data record 128 has a length of 76 bytes.

It contains module parameters as well as channel or technology parameters of the 32 channels; 2 bytes each.

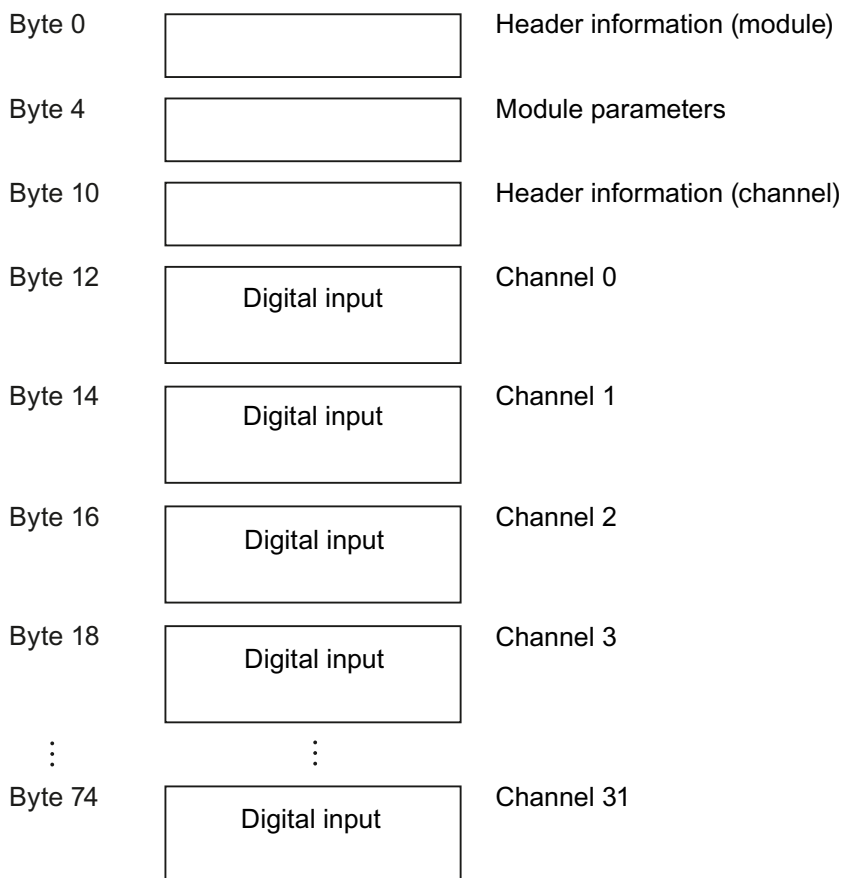


Figure A-1 Structure of data record 128

Header information and module parameters

The following figure shows the configuration of the header information and module parameters.

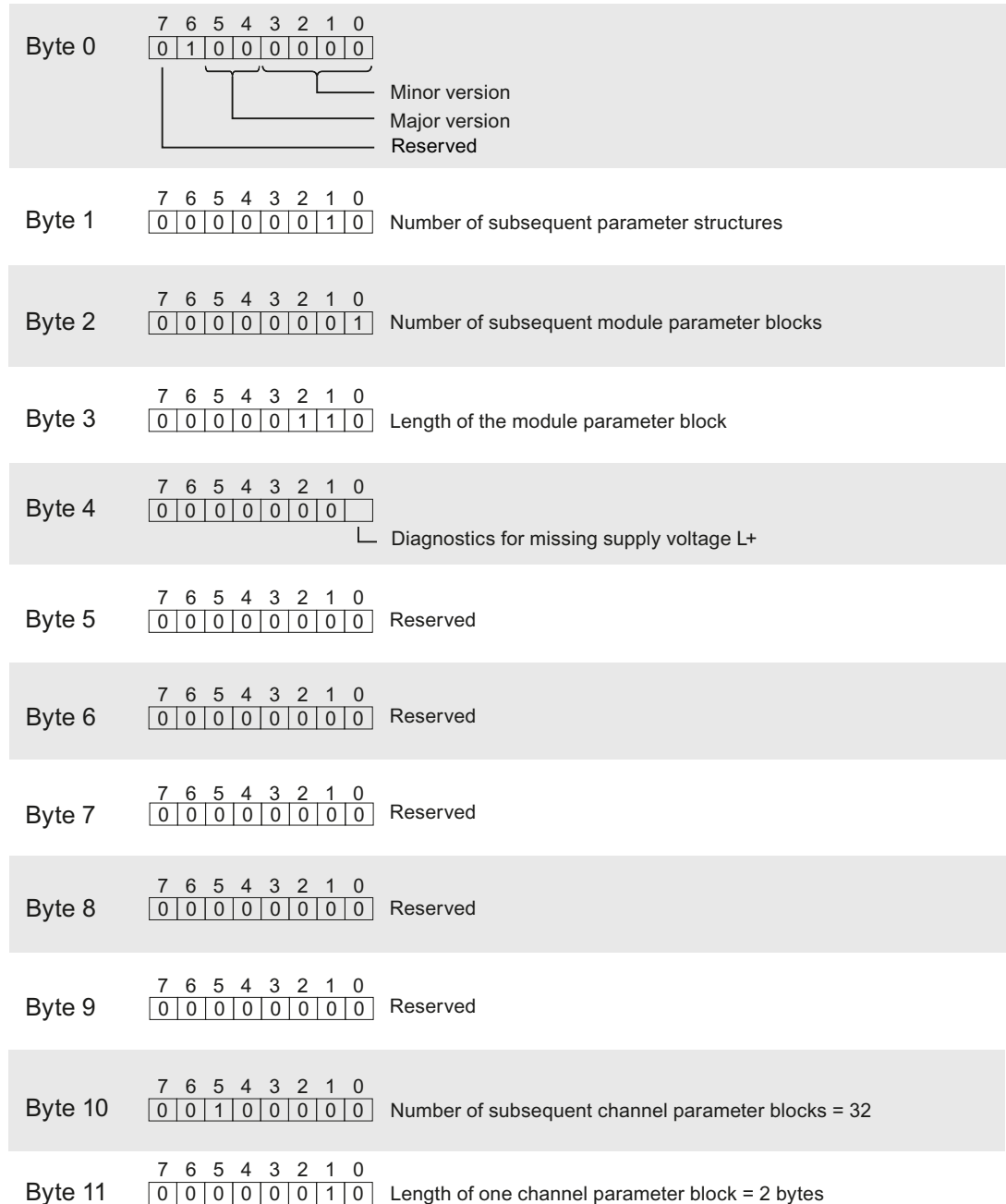


Figure A-2 Header information and module parameters of data record 128

Channel parameters

Every channel parameter block contains the channel parameters for the digital input (2 bytes).

The following figure shows the configuration of the channel parameters for channels 0 to 31.

$x = 12 + (\text{channel number} * 2)$; with channel number = 0...31

All unused bits and the bits or bytes marked as "reserved" must be set to zero.

You activate a channel parameter by setting the corresponding bit to "1" or the corresponding value.

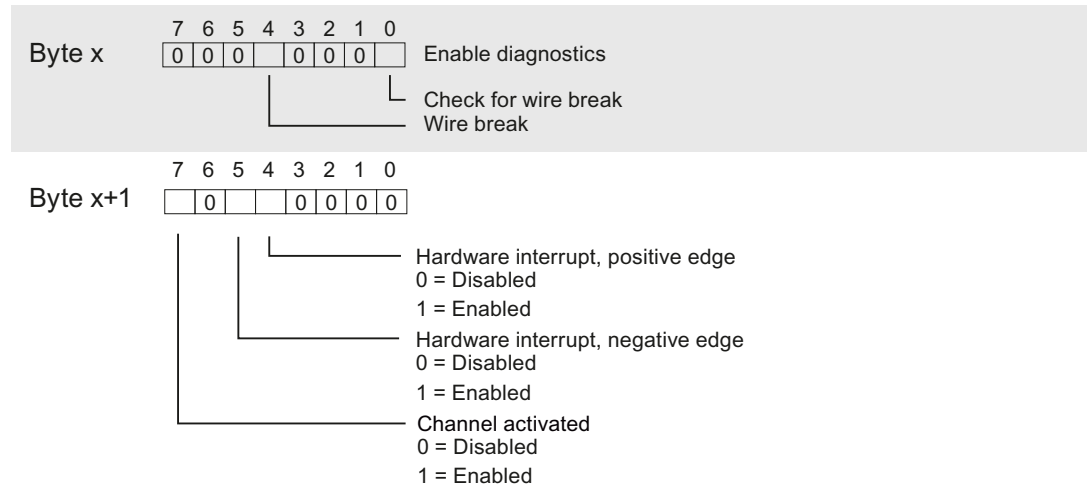


Figure A-3 Structure of bytes x to x+1 for channels 0 to 31

A.2.3 Diagnostics messages and maintenance events

Diagnostics messages

A diagnostics message is generated for each diagnostics event and the DIAG LED on the I/O module flashes red or the MT LED lights up yellow.

The diagnostics messages can be read in the diagnostics buffer of the CPU, for example.

Diagnostic messages are assigned either channel-specific to an input or assigned as module messages to all inputs. All channels are switched off for diagnostics messages that affect the

entire module. Only the corresponding digital input is affected by diagnostics messages, which refer to individual channels.

Table A-1 Diagnostic messages, their meanings and remedy options

Diagnostics message	Error code	Assignment	Meaning	Remedy
Wire break	6 _H	Channel	Impedance of encoder circuitry is too high	Use another type of encoder or wiring, for example, use conductors with a larger cross-section
			Wire break between the module and sensor	Connect the cable
			Channel not connected (open)	• Disable diagnostics • Connect encoder contact with a 15 kΩ to 18 kΩ resistor. • Switch off wire break check
Parameter assignment error	10 _H	Module	• Parameter assignment does not match the terminal block used • Incorrect parameter assignment	• Correct the parameter assignment • Check terminal block
Supply voltage missing	11 _H	Module	Missing or insufficient supply voltage L+	• Check the supply voltage L+ on the terminal block • Check terminal block
Hardware interrupt lost	16 _H	Module	Error is reported because the events occur faster than they can be processed in the system.	Check the system and remedy the cause.
Channel/component temporarily unavailable	1F _H	Module	Firmware update is in progress or has been canceled during the update. The module does not read any process values in this state.	• Restart the firmware update • Wait for firmware update
Module is faulty	100 _H	Module	I/O module is faulty	Replace I/O module
Retentive memory in carrier module defective	154 _H	Module	An error of the memory block on the adapter module was detected during startup	Replace carrier module
Retentive memory in the terminal block defective	155 _H	Module	An error of the memory block on the terminal block was detected during startup	Replace terminal block

Maintenance events

A maintenance event is generated for each maintenance required identified. The MT LED lights up on the module.

Maintenance messages do not have a direct influence on the functions of the module or digital inputs.

Table A-2 Maintenance messages, their meaning and possible remedies

Maintenance message	Error code	Assignment	Meaning / Cause	Remedy
Retentive memory in carrier module defective	154 _H	Module	Fault detected in the data block on the carrier module during operation	Replace carrier module
Retentive memory in the terminal block defective	155 _H	Module	An error of the memory block on the terminal block was detected during operation	Replace terminal block

A.2.4 Hardware interrupts

Hardware interrupts

The DI 32x24VDC HA generates a hardware interrupt for the following events:

- At a rising edge (signal change from 0 to 1)
- At a falling edge (signal change from 1 to 0)

Interrupt OBs are called automatically if an interrupt occurs.

You can find detailed information on the event in the hardware interrupt organization block with the "RALRM" instruction (read additional interrupt information).

Blocks OB40 to OB47 are possible hardware interrupt organization blocks.

The module channel that triggered the hardware interrupt is recorded in the start information of the hardware interrupt organization block in the OB-specific tag (for example, OB40_POINT_ADDR).

The figure below shows the assignment to the bits of local data double word 8.

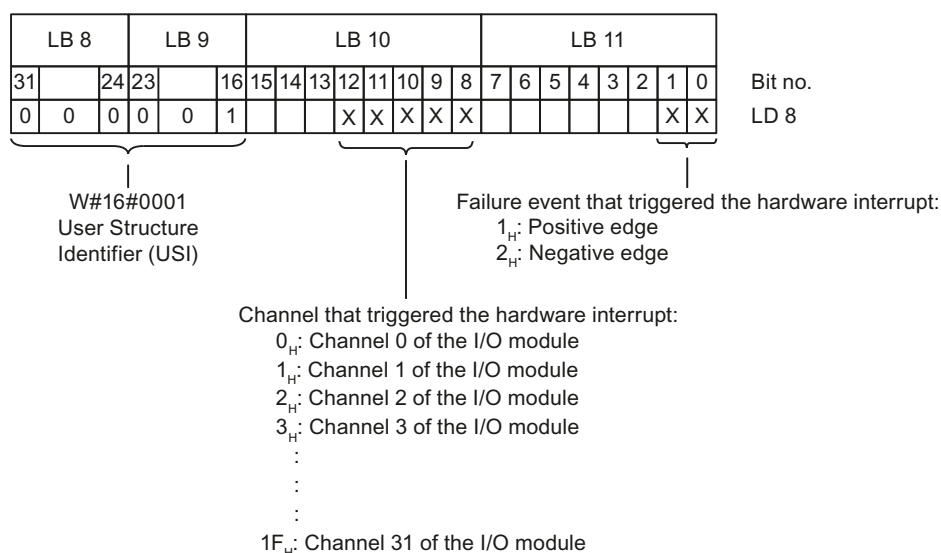


Figure A-4 Start information of the organization block

A.2.5 Address space

Abbreviations

- "IB" stands for input byte, that is, the module start address in the input area
- "DI_n" stands for digital input n
- "QDI_n" stands for the value status (QI) of digital input n

Address space

The following figure shows the allocation of the address space of the DI 32x24VDC HA.

Table A-3 Address space of the DI 32x24VDC HA

IB x +	7	6	5	4	3	2	1	0
0	DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0
1	DI15	DI14	DI13	DI12	DI11	DI10	DI9	DI8
2	DI23	DI22	DI21	DI20	DI19	DI18	DI17	DI16
3	DI31	DI30	DI29	DI28	DI27	DI26	DI25	DI24
4	QDI7	QDI6	QDI5	QDI4	QDI3	QDI2	QDI1	QDI0
5	QDI15	QDI14	QDI13	QDI12	QDI11	QDI10	QDI9	QDI8
6	QDI23	QDI22	QDI21	QDI20	QDI19	QDI18	QDI17	QDI16
7	QDI31	QDI30	QDI29	QDI28	QDI27	QDI26	QDI25	QDI24

Evaluate value status

There is a one bit value status for each digital input in the input address space.

Irrespective of the diagnostics enables, each value status provides information on the validity of the corresponding process value.

- Value status = 1: Process value okay, "good"
- Value status = 0: Process value not okay, "bad"

Basically, the value status is set to "good" if the digital value can be detected without error. The value status is set to "bad" in the following cases:

- The digital value cannot be detected due to an error.
- The digital channel is deactivated.

Note

The value status is not set to "bad" when there is a wire break on the channel and the wire break test is disabled.

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